Notes on
Gabbanelli, Drazer & Koplik Article

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NB: Notes for Chapter 2 from Gabbanelli, Drazer and Koplik's Lattice Boltzmann method for non-Newtonian (power-law) fluids. Anything in bold purple has a footnote with a question.

LATTICE BOLTZMANN METHOD

The LBM can be viewed as an implementation of the Boltzmann equation on a discrete lattice and set of velocity distribution functions,

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta x, t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i(f(\mathbf{x}, t)) \tag{1}$$

where

- f_i is the particle velocity distribution in the i^{th} direction.
- $\Omega_i(f(\mathbf{x}, t))$ is the collision operator.
- $\Delta x, \Delta t$ are the space and time discretisations.

From this, the density ρ and momentum density $\rho\mathbf{u}$ can be found in the usual way,

$$\rho = \sum_i f_i, \quad \rho\mathbf{u} = \sum_i f_i \mathbf{e}_i \tag{2}$$

and for here on we are concerned with incompressible flows.

Bhatnagar-Gross-Krook approximation

Assuming the system is close to **equilibrium**[1] with local equilibrium function f_i^{eq}, then the BGK approximation of the LBM is

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta x, t + \Delta t) = f_i(\mathbf{x}, t) + \frac{f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)}{\tau} \tag{3}$$

where τ is the relaxation time and is related to the kinematic viscosity ν as $\nu = (2\tau - 1)/6$.

[1] What this mean? Is this a safe assumption?

Non-Newtonian Flows

The idea is that we use a different relaxation time τ at each node, turning the global relaxation time into a local **variable**[2].

The generalised non-Newtonian model gives the stress tensor σ_{ij} in terms of the rate-of-strain tensor D_{ij} as $\sigma_{ij} = 2\mu D_{ij}$ where $\mu = \mu(D_{ij})$ is a function of the invariants of the tensor D_{ij}.

Using the power-law model which gives $\mu = m\dot{\gamma}^{n-1}$ with $\dot{\gamma}$ the magnitude of the local shear rate related to D_{ij} as $\dot{\gamma} = \sqrt{D_{ij}D_{ij}}$.

The issue with this is that for zero shear rates the effective viscosity diverges and for shear-thickening fluids at zero shear rates the viscosity vanishes.

Proposed is the truncated power-law,

$$\nu(\dot{\gamma}) = \mu(\dot{\gamma})/\rho = \begin{cases} m\dot{\gamma}_0^{n-1}, & \dot{\gamma} < \dot{\gamma}_0 \\ m\dot{\gamma}^{n-1}, & \dot{\gamma}_0 < \dot{\gamma} < \dot{\gamma}_\infty \\ m\dot{\gamma}_\infty^{n-1}, & \dot{\gamma} > \dot{\gamma}_\infty \end{cases} \tag{4}$$

where the lower and upper values of the kinematic viscosity are $\nu_{\text{min}} = 0.001$ and $\nu_{\text{max}} = 0.1$. This overcomes the issue by breaking up the shear rates into three regions, each region well behaved and makes physical sense.

[2] I think this is the gist but I am not too sure.